- The linked list could be a valid option if removing old entries was more important than storing entries or finding cache elements: but in that case, the cache would basically be useless, and adding it would provide no benefit.
- In all cases, the memory needed to store `n` entries is `O(n)`.

Now the question is: can we do any better?

7.4.1 Sometimes you have to double down on problems

As you can imagine, if the answer was no, we wouldn't really be asking the question in the first place.

But how can we do any better? There isn't any other data structure that allows us to optimize all three main operations at the same time.

And yet… What if I told you that it is possible to design an LRU cache that takes `O(1)` amortized time[19] for all those operations?

Before you go on and read the solution to the riddle, please try for a couple of minutes to imagine how that could be possible.

Let me give you two hints (spoiler alert: you might want to try designing a solution before reading them):